

ECE 486: Lab 5 Report

UNIVERSITY OF
Waterloo



Department of Electrical and Computer Engineering

A Report Prepared For:

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ECE486

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Introduction

In Lab 5, we completed the task Option 2 (M-O) from the lab manual. The objective of Lab 5 is to develop a vision-based robotic grabbing system, which is able to visually detect building blocks in the workspace, control the robot to grab them, and place them at the position with desired ArUco marker. Although the lab manual instructed us to use gripper, the actual block size was larger than the gripper's diameter. Therefore, under the guidance of lab instructor Dr. Cooper, we used a pneumatic suction cup instead.

Strategy

Before each experiment, we repeated the camera calibration several times using the method in previous labs, until the camera matrix and distortion coefficient never contain outliers or extreme elements, and then re-calculate the transformation matrix and translation vector. This will help reduce positioning errors accumulated from previous experiments. Accurate calibration is essential for minimizing camera-to-robot errors.

Camera matrix:

$$C = \begin{bmatrix} 942.06711744 & 0 & 278.79611215 \\ 0 & 954.98658013 & 268.68056973 \\ 0 & 0 & 1 \end{bmatrix}$$

Distortion coefficient:

$$D = \begin{bmatrix} -0.03197389 \\ 0.61666864 \\ 0.00632412 \\ -0.00394646 \\ -2.95634379 \end{bmatrix}$$

Transformation matrix:

$$R = \begin{bmatrix} -0.97389296 & -0.09716424 & 0.2051624 \\ -0.13530187 & 0.97414712 & -0.18091656 \\ -0.18227975 & -0.20395222 & -0.96186152 \end{bmatrix}$$

Translation vector:

$$T = \begin{bmatrix} 0.18643354 \\ 0.16327346 \\ 0.43956406 \end{bmatrix}$$

The building blocks on the table were detected using color segmentation in the HSV color space. Since the workspace on the table is primarily white, and the building blocks are bright red or blue, this method provides reliable results after adjustment, with a saturation filter set from 150 to 225. In addition, we also detected the area of the color contour. If the area was less than 45 pixels, we would not consider it as a building block. This helps to eliminate the false positive from the image

noise. Morphological filtering is applied to remove small noise areas, and contours are extracted to determine the center of each foreign object.

The center of each detected building block was obtained in pixel coordinates. The pixel coordinate was transformed to robot coordinate using those calibrated camera parameters. Given the pixel coordinate $\mathbf{p} = \begin{bmatrix} x_p \\ y_p \end{bmatrix}$, the corresponding undistorted image coordinate $\mathbf{p}_\tau = \begin{bmatrix} x_{\tau p} \\ y_{\tau p} \end{bmatrix}$ can be obtained by python OpenCV Cv2 package, and the robot coordinate can be calculated as

$$\mathbf{x} = \mathbf{T} + \lambda R \begin{bmatrix} x_{\tau p} \\ y_{\tau p} \\ 1 \end{bmatrix}$$

$$\lambda = \frac{z_g - T_z}{d_z}$$

where $\mathbf{T} = \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix}$, $\mathbf{d} = \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix} = R \begin{bmatrix} x_{\tau p} \\ y_{\tau p} \\ 1 \end{bmatrix}$, and z_g is the suitable height for robot to grab the building

block on the table, and according to the measurement, $z_g = -45$ mm. We also defined a safe moving height $z_s = 50$ mm, which is the safe height for the robot the move horizontally without hit the building blocks on the table.

In addition, there are three ArUco markers on the table, and we need to choose one as the position that the building blocked will be placed. We are asked to enter the number of the ArUco marker we want. Calculating the coordinates of the ArUco marker can be done use part of the code from the previous lab, which will automatically identify the position of the ArUco marker, measure the camera coordinate $\mathbf{p}_m$, and then calculate the robot coordinate as $\mathbf{x}_m = R\mathbf{p}_m + \mathbf{T}$.

After all the coordinates was obtained, the robot first moved to a predefined safe height z_s , and moves horizontally until it reaches the top of the building block, then descends vertically to the grabbing height z_g , where the suction cup was activated to pick up the object. After grabbing, the robot lifted the object back to z_s , moves horizontally to the position above the ArUco marker, descends back to grabbing height z_g , deactivated the suction cup to put building block down, and finally returned to the home position.

The task was considered successful if the robot can correctly detect, grab, and place the building block to the selected ArUco marker without dropping or colliding with other objects. In our code, after one building block was placed, you can choose to press Enter to continue detect and move the next building block, if there is not only one block in the workspace, or press q to terminate the program.

Experimental Design

To evaluate the robotic arm's visual detection and grabbing performance, we conducted multiple experiments. In each experiment, a random ArUco marker from 3 available markers was selected as target position, and the building blocks were randomly placed in the workspace, with random positions and orientations. Throughout the experiments, we ensured that the camera and robot

base remained stationary, any accidental movement required recalibration. In each experiment, we observed whether the block was successfully detected, grabbed, and placed at the selected ArUco marker without collisions or dropping during movement.

Results

During our test, the robot successfully detected the building blocks, and used a pneumatic suction cup to successfully grab it and place it to desired target. Due to errors in camera calibration or mechanical issue, we needed to adjust the offset parameters in the x or y directions to eliminate some regular deviations. Finally, although a small amount of positioning error remained, the robot was able to reliably complete the required task.

Reflection and Future Improvement

In our experiment, although the camera had been calibrated multiple times, subsequent adjustments to the offset parameters were still necessary to counteract some deviations, which may have originated from camera calibration errors. Our current camera calibration was merely to check for outliers in the camera matrix. However, after calibrating the camera, we could actually reuse the code in Lab 3, which pick 20 points in the workspace to test and calculate the average error between desired robot coordinate and camera calculated robot coordinate, thereby testing the accuracy of the camera calibration.

Furthermore, robotic arm injuries are common in industry, so safety is a critical concern. In this lab, we need to manually place blocks to test the robotic arm's functionality, requiring it to remain stationary during this process. Currently, we require the robotic arm to only move after the operator moves their hand and presses Enter, with the message "Please remove your hand" continuously printed on the screen. However, this method still needs improvement. When I participated in the Toyota Innovation Challenge, one participant wrote code that controlled a robotic arm by recognizing human gestures; this code could be implemented using Python's MediaPipe Hands. We could write similar code to automatically stop the robotic arm when a human hand is detected by the camera.

Another improvement would be to modify the motion planning according to the selected end-effector. The original program was developed for a pneumatic gripper, so the orientation of the building block was considered. In our original program, the robot will rotate the end effector until the gripper is orthogonal to the long side of the detected building block before grabbing it. However, under the instruction of Dr. Cooper, we used a pneumatic suction cup rather than a pneumatic gripper during the real experiment. With a pneumatic suction cup, the end effector rotation was no longer required. However, this rotation program was still executed in our code, causing the suction cup to rotate in place before grabbing. This extra rotation not only increased the grabbing time, but also twisted the pneumatic tubing during our real experiment and may cause damage of the tube and reduce the reliability of the robot arm. Removing this extra rotation program would improve both efficiency and operational safety.

Extra Information kindly for Dr. Cooper:

My lab partner has decided to withdraw from the term before Lab 4, so I have to complete all the tasks of 2 labs by myself. The professor Dr. Gennaro Notomista is aware of this and has indicated that extra grade curve may be considered on me if my lab report and project report quality is not very good, depending on the circumstances. Thanks for consideration.